

Design and Implementation of an Intelligent Autonomous Railway Crossing Safety Framework with Multi-Sensor Redundancy

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Abstract—Conventional railway level crossings suffer from high accident rates due to delayed gate operations, single-point sensor failures, and lack of remote telemetry. This paper presents an intelligent, autonomous railway crossing safety framework utilizing multi-sensor redundancy and real-time telemetry powered by an ESP32 microcontroller. A redundant array of four heterogeneous sensors—two ultrasonic and two infrared (IR)—is deployed across the left and right approaches of the track. A "3-out-of-4" majority voting logic combined with a temporal verification algorithm (1000 ms fault-confirmation window) handles sensor degradation: if a single sensor fails to detect an approaching train while the remaining three concur, the framework latches a fault condition, flags the damaged node, and generates a diagnostic warning. Real-time telemetry, including sensor distance vectors, detection states, gate positioning, and fault diagnostics, is streamed via a 5 Hz Bluetooth Serial link to a mobile supervisory application. Dual servo-actuated barriers, synchronized with auditory buzzers and visual LED indicators, provide immediate physical containment. Empirical validation demonstrates absolute fault tolerance with zero system downtime during simulated single-sensor failures, defensive gate closure within 20 ms of detection, and fault latching within exactly 1000 ms, confirming that the proposed redundancy layer eliminates single-point-of-failure vulnerabilities in autonomous level crossings.

Index Terms—Railway level crossing, sensor redundancy, majority voting, fault-tolerant systems, ESP32, real-time telemetry, Bluetooth, embedded systems.

I. Introduction

Railway level crossings (LCs) represent the intersecting boundaries where railway tracks cross roads at the same level and constitute one of the most hazardous zones within global railway infrastructure. Traditional manual gate operations are prone to human delays, fatigue, and miscommunication between station masters and gatekeepers. Automated systems using a single set of sensors exist, but they present a significant hazard: if the single sensor tracking the train fails due to environmental factors such as dirt, rain, fog, or physical misalignment, the system fails to close the gates, leaving the crossing open to vehicular traffic while a train approaches.

This work addresses these safety gaps by developing an intelligent, autonomous, and highly reliable infrastructure framework. By shifting from vulnerable single-sensor configurations to a multi-sensor redundant architecture, the framework ensures that environmental or electrical degradation of a single component does not compromise public safety. Integrating real-time telemetry via Bluetooth to a mobile

application further shifts maintenance from a reactive to a proactive, predictive model, minimizing downtime and reducing human error.

A. Related Work

Single-sensor automation schemes rely purely on a single IR or ultrasonic sensor to log train arrival; while faster than manual intervention, they enter a blind state once obstructed by debris [1]. Homogeneous multi-sensor architectures introduce redundancy using multiple sensors of the same type, but remain vulnerable to common-cause failures such as fog or rain disrupting multiple ultrasonic units simultaneously [2]. Advanced IoT and long-range monitoring systems integrate GSM/Wi-Fi telemetry to central servers, but typically treat telemetry as a passive logging function rather than an active, localized diagnostic tool, and lack edge-computing logic to isolate a specific faulty sensor while a train is actively passing [3], [4].

The identified research gaps are: (i) absence of integrated, real-time edge diagnostics capable of isolating specific sensor failures during an active detection cycle; (ii) reliance on single-medium sensor arrays vulnerable to environmental blind spots; and (iii) lack of a low-latency local telemetry standard streaming raw distance metrics alongside structured fault arrays. This work bridges these gaps using an ESP32-WROOM-32 platform that combines heterogeneous sensors (ultrasonic and infrared) into a localized 3-out-of-4 majority voting matrix, creating an edge-computing safety device that both automates gate closure and actively monitors its own hardware health.

B. Objectives

1) Fault-Tolerant, Multi-Sensor Edge Detection: Design and program a heterogeneous sensor array (dual ultrasonic, dual infrared) managed by a 3-out-of-4 voting algorithm that triggers protective gate closure (90° servo sweep, buzzer, red LED) when at least one sensor detects a train under normal conditions, or when at least three sensors concur during a single-sensor failure state.

2) Real-Time Telemetry and Edge Diagnostics: Engineer a data-streaming pipeline that determines hardware fault conditions via a 1000 ms confirmation window and packages sensor and fault data into CSV telemetry strings transmitted at 5 Hz over Bluetooth Serial to a mobile application for remote monitoring.

II. Methodology

The methodology leverages concurrent hardware interfacing and real-time software evaluation on an ESP32 microcontroller,

split into three layers: physical input sensing, edge processing/decision, and output actuation/telemetry.

A. System Model

The physical layout simulates a two-way track approach. The Left Track Zone is guarded by Ultrasonic Sensor 1 (US1) and Infrared Sensor 1 (IR1); the Right Track Zone is guarded by Ultrasonic Sensor 2 (US2) and Infrared Sensor 2 (IR2). The sensors operate concurrently: ultrasonic sensors track physical distance bounds via echo-return timing, while IR sensors provide fast line-of-sight digital logic shifts. Inputs feed directly into the ESP32, where input-only, 5V-tolerant pins (GPIO 34, 35, 36) receive signals without pull-up interference, while high-speed digital pins drive the servo motors, buzzer, and indicators.

B. Algorithm and Flow Control

The system executes a strict path inside an infinite loop: sensors are read, detections evaluated, and a detection count computed. If the count is zero, all latched faults and timers are cleared. For each sensor, if the detection count equals three (majority concurrence) while that specific sensor reads clear, a 1000 ms confirmation timer starts; if this persists beyond the window, a fault is latched for that node. Train presence is determined by detection count > 0 OR any latched fault, triggering gate closure, red LED, and buzzer; otherwise the gate opens and the green LED activates. Telemetry is packaged and streamed via Bluetooth every 200 ms, and the loop ticks at a 20 ms interval. Fig. 1 illustrates this control flow.

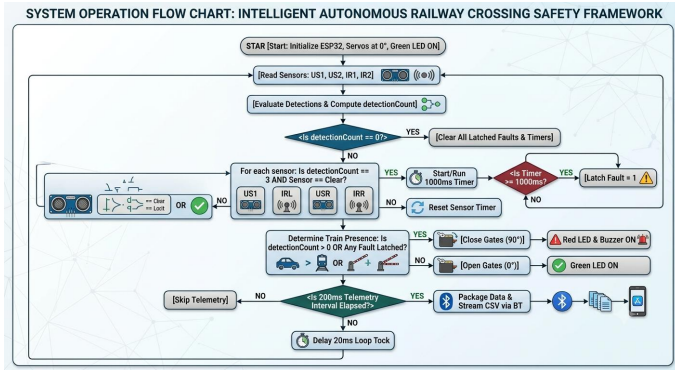


Fig. 1. System operation flow chart of the intelligent autonomous railway crossing safety framework.

C. Mobile Supervisory Application

A companion Android application, developed in MIT App Inventor, connects to the ESP32 over Bluetooth Serial. On a timer tick, it checks for available bytes, splits the received CSV string, and updates on-screen labels for gate status and each sensor's live reading; fault indicator buttons are recolored red when the corresponding fault flag equals one. Fig. 2 shows the block design, and Fig. 4 shows the resulting interface, which displays gate status alongside individual sensor and fault indicators for US Left, IR Left, IR Right, and US Right.

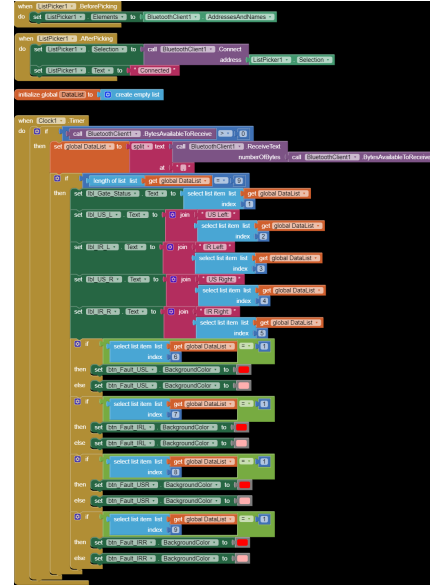


Fig. 2. Block design of the mobile monitoring application (MIT App Inventor).

D. Hardware Implementation

Fig. 3 shows the simulated wiring diagram: dual HC-SR04 ultrasonic sensors, dual IR modules, an ESP32-WROOM-32, dual servo barriers, status LEDs, and a piezo buzzer. Table I lists the calibrated hardware and software operating parameters.

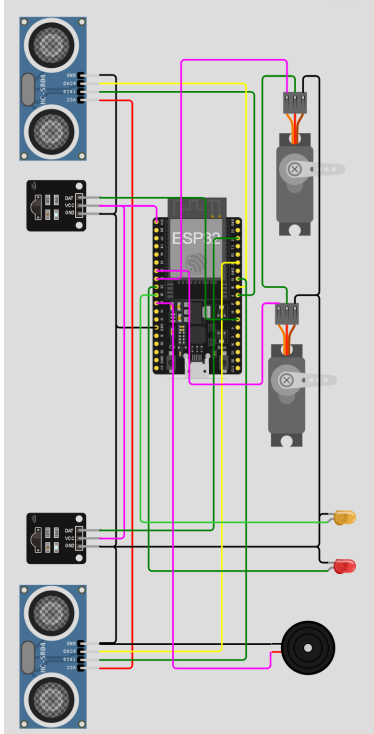


Fig. 3. Simulation wiring diagram of the sensor-actuator-controller assembly.

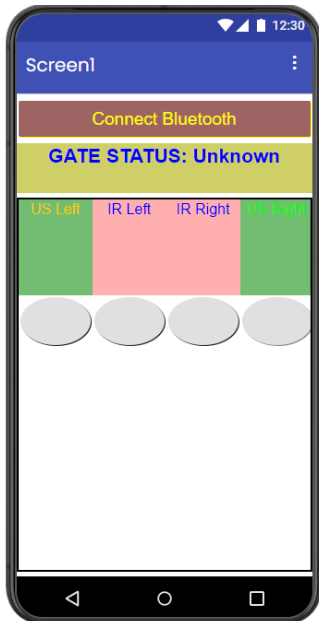


Fig. 4. Mobile supervisory application interface.

TABLE I
OPERATIONAL HARDWARE AND SOFTWARE SETTINGS

Parameter	Value	Purpose
trigPin1, trigPin2	5, 19	10 μ s sonic pulse generation
echoPin1, echoPin2	34, 35	Echo duration input (safe, input-only)
irPin1, irPin2	22, 36	IR digital proximity logic
servoPin1, servoPin2	32, 33	PWM channels for gate barriers
Distance threshold	10.0 cm	Max. ultrasonic trigger distance
FAULT_CONFIRM_MS	1000 ms	Fault confirmation window
txInterval	200 ms	Bluetooth telemetry interval (5 Hz)
Pulse timeout	30000 μ s	pulseIn() upper bound

III. Results and Discussion

The framework was tested across multiple simulated train transits to assess speed, precision, and stability under both normal and fault conditions.

When a train enters the detection zone under nominal conditions, all four sensors trigger simultaneously (detectionCount = 4). The system switches states in under 20 ms, driving both servo barriers from 0° (open) to 90° (closed), activating the red warning LED, and sounding the buzzer, as shown in Fig. 5.

The primary innovation is observed under sensor degradation. During testing, one sensor was purposefully covered or disconnected before a train run. When the simulated train arrived, three sensors registered the train while the faulty sensor reported clear (detectionCount = 3). Rather than fluctuating or failing to protect the crossing, the system: (a) immediately initiated a protective gate closure because detectionCount > 0; (b) started a millisecond timer for the non-responsive sensor; and (c) latched a permanent fault flag once the timer crossed 1000 ms. This latched fault ensures the gate remains safely down until the train fully clears and detectionCount returns to 0, as shown in Fig. 6.

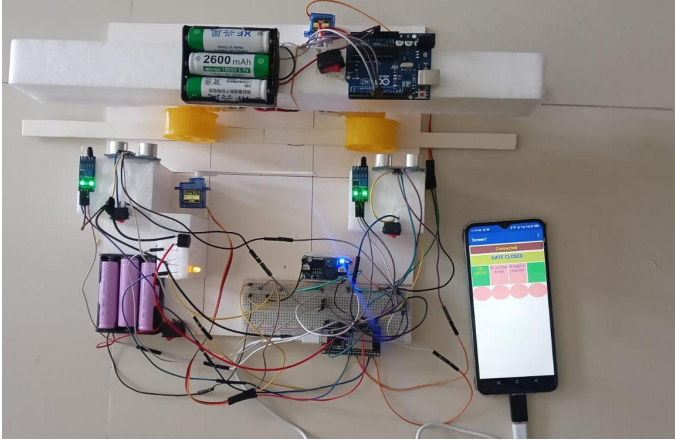


Fig. 5. System output when a train is detected and the gate is closed.

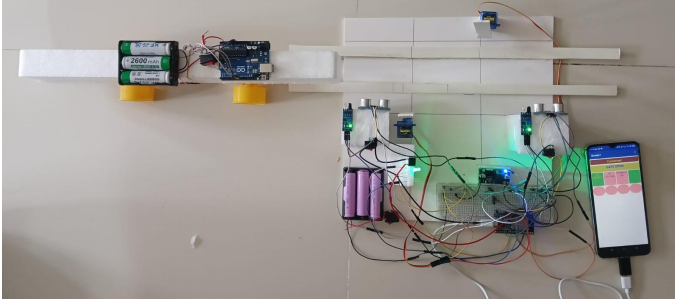


Fig. 6. System output after the train has departed and the gate re-opens.

Table II summarizes the empirical telemetry captured over the Bluetooth link across five representative operating states, from an idle track through nominal detection, fault-window onset, fault latching, and reset.

TABLE II
SYSTEM RESPONSE AND TELEMETRY MATRIX

State	Sensors (US1,IR1,US2,IR2)	Count	Fault Vector	Gate	Latency
Idle	Clear,Clear,Clear,Clear	0	0,0,0,0	Open	<5 ms
Nominal entry	Det,Det,Det,Det	4	0,0,0,0	Closed	<20 ms
Fault window	Clear,Det,Det,Det	3	0,0,0,0	Closed	1 ms (timer starts)
Fault latched	Clear,Det,Det,Det	3	1,0,0,0	Closed	≥1000 ms
Reset	Clear,Clear,Clear,Clear	0	0,0,0,0	Open	<20 ms

The telemetry system ran reliably at its 200 ms pacing interval, consuming minimal processing overhead while keeping

data streams clean and legible. The 30 ms timeout on the ultrasonic pulseIn() function successfully prevented software lockups, ensuring that a missing echo returned a clean -1 value without stalling the program loop. These results validate the system's design, confirming that it provides a reliable, self-monitoring solution for autonomous railway safety.

IV. Conclusion and Future Scope

The 3-out-of-4 voting algorithm was successfully implemented on the ESP32, and testing confirmed that the system responds quickly to train detection by activating the gate, warning LEDs, and buzzer, continuing to operate safely even during single-sensor failure scenarios. The real-time monitoring system successfully detected and reported sensor faults, with fault information transmitted to the mobile application via Bluetooth at a 5 Hz update rate, enabling effective remote monitoring. Future work will integrate LTE-M, NB-IoT, or LoRaWAN communication for long-range remote monitoring, add solar power and battery management for off-grid deployment, incorporate camera-based computer vision to detect vehicles trapped on the crossing, and implement predictive maintenance algorithms to identify sensor degradation before failure occurs.

References

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